NAME:KRITHIKA.S

SRN:PES2UG24CS238

CLASS:4D

Github link: [krithika297/PES2UG24CS238-pes-vcs]

1a

```
gcc -o test_objects test_objects.o object.o -lcrypto
Stored blob with hash: d58213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9d
b5e5c4b9aafc12
Object stored at: .pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed
0221bbe9db5e5c4b9aafc12
PASS: blob storage
PASS: deduplication
PASS: integrity check

All Phase 1 tests passed.
krithika_s@88U9592OG990I:~/PES2UG24CS238-pes-vcs$
```

1b

```
krithika_s@88U9592OG990I:~/PES2UG24CS238-pes-vcs$ find .pes/objects -type f
.pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
.pes/objects/2a/594d39232787fba8eb7287418aec99c8fc2ecdaf5aaf2e650eda471e566fcf
.pes/objects/25/ef1fa07ea68a52f800dc80756ee6b7ae34b337afedb9b46a1af8e11ec4f476
krithika_s@88U9592OG990I:~/PES2UG24CS238-pes-vcs$
```

2a

```
krithika_s@88U9592OG990I:~/PES2UG24CS238-pes-vcs$ make test_tree
./test_tree
gcc -Wall -Wextra -O2 -c tree.c -o tree.o
tree.c: In function 'tree_from_index':
tree.c:167:13: warning: implicit declaration of function 'object_write' [-Wimplicit-function-declaration]
  167 |         if (object_write(OBJ_BLOB, data, size, &te->hash) != 0) {
      |             ^~~~~~~~~~~~
tree.c:163:9: warning: ignoring return value of 'fread' declared with attribute 'warn_unused_result' [-Wunused-result]
  163 |         fread(data, 1, size, f);
      |         ^~~~~~~~~~~~~~~~~~~~~~~
gcc -o test_tree test_tree.o object.o tree.o -lcrypto
Serialized tree: 139 bytes
PASS: tree serialize/parse roundtrip
PASS: tree deterministic serialization

All Phase 2 tests passed.
krithika_s@88U9592OG990I:~/PES2UG24CS238-pes-vcs$
```

2b

```
krithika_s@88U9592OG990I:~/PES2UG24CS238-pes-vcs$ find .pes/objects -type f
.pes/objects/65/df4fd62f3b96e93f33b3b3924c21ebe5331bf54770d8fa7c9a5018efaf3008
.pes/objects/e2/d8026a766be394788e227b71b9fb17cfeb33c2966577742770d2ca9e977028
.pes/objects/05/c8d777d0be85748b6788d1b5caf7d69754e6e68060c265ca21cbd6c05e5f43
.pes/objects/0d/36b699f2fb606b6af430b611f65c351183b9ea05c0bf82b7c5cebb6d4899b0
.pes/objects/c9/ed899783ff70cac61dbbc4e205d04714402c853c4e5f6107d2c8b21cd3a8f4
.pes/objects/a0/9cd29b4a4df5ae81a1da9f06d60998bf142b62356158f9d991366b92fe90e5
.pes/objects/6b/1a89bbbd6a604d675a2fa7e4d59aa0d38e3867805722b60e37aa21bb195414
.pes/objects/62/19dea03b95f88014daaf8d0c1f2c3876a4df7e0dd8f7bf4a2b8f9c4025f4e3
.pes/objects/62/27d26df b567be18fd00cb3d22f14359e1e658f05659ea73b19aa54fcff0614
.pes/objects/a7/4c8d33a8cdae4949c89c9996cf1d246bbd04c40826d9dc7d25fbe611236b73
.pes/objects/60/459622a6a99c40ddfcd668e7cbd0d70a4d53aeb448a251aa982ea2043e3838
.pes/objects/b9/9f25b5b7ce8abc32c8255ce7657c1a9f0f68e48f24f2ef9bcb9cda94bac19f
.pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
.pes/objects/48/1407f7237f7bd749dca31e2d8422bafe211741d5556f007e0faf821ffc7179
.pes/objects/2a/594d39232787fba8eb7287418aec99c8fc2ecdaf5aaf2e650eda471e566fcf
.pes/objects/66/224663d23e6f4d9de9e2c7e6d8764305a92a3830a1a52d3d5f4aa8007b5c39
.pes/objects/fa/a8239e9965524e41104d2e01c4699a169a858a65dbd76f18feb21ea9459090
.pes/objects/4e/00bf1c79aeace2a31aeb33fe800af4a5667b8e3ae9ecb63ab27dc6e3f90851
.pes/objects/94/4f06dc83e3ca6e0f4573268f03de16d5a3dfd2cc43165878fe1d3e6972ee35
.pes/objects/e1/8f77c5c81b723ce4688bc7e2dc8a398bfa2501649445a8d1a21bacad85fbd9
.pes/objects/8e/128fab627eae587dffd5b1bf4bc8e5a7a75af7c3aac302fc408cb35f022b5a
.pes/objects/c6/53a193f484b05908d36c4c314a6f767241aee29be344d2abaf62de28958d9d
.pes/objects/ac/b78445564c5ed9d182330bea8fc027304964e15914deaceb7bb22618fe35ff
.pes/objects/c7/727bb3a915f59a2c4543394625c111bfb925e7975d4fa9339bc24377bd9c38
.pes/objects/25/ef1fa07ea68a52f800dc80756ee6b7ae34b337afedb9b46a1af8e11ec4f476
.pes/objects/10/d488e897a9291aca9f7f55db526cf1c0a4d2a6274de8d6729d8da036cce9fd
.pes/objects/8c/66283a8bc77d49cd1603623f437bdd1d363a4bbf5cd5f13e10d71669c592d8
.pes/objects/8f/c13b3ff597c5d58c22d59cc057f3f2d5d4becafd8578835da61dbd0dd4e0b3
.pes/objects/e9/5829cbebfc4ff105b206ece482623e221adfa1973ba6a858531b6d74739708
.pes/objects/5d/f21d2cc4be79af197e44167eb131688d98f9c77dae5aba8e381a20512027d9
.pes/objects/43/72fc14cf3990ffbc0d87e98d84d4cd14e07bf8547caca89d8c9f7f8f3103f4
.pes/objects/f7/7c03788f0e18ff8b43c8134f449b8f19b890cc74faf3329e4c9bb978aefc03
.pes/objects/e0/0c50e16a2df38f8d6bf809e181ad0248da6e6719f35f9f7e65d6f606199f7f
.pes/objects/a1/6c9a05468adae6e754a3f5bf49533153c3b0f32c2444b615c142e18fbc8e35
.pes/objects/34/3d34bf690fefa64dc0d95a01280df0e5d489c8eced213340be03ea5ffacc2b
.pes/objects/2c/f8d83d9ee29543b34a87727421fdecb7e3f3a183d337639025de576db9ebb4
.pes/objects/30/1784349754fedc835be3ca066dbfd5c16cb1ffbf6539e86be514ba37fe6bf7
.pes/objects/6e/2ece108ec707b076dc85e5691d8d82601dc9d5e4b275d26d30eadb8ccdda81
.pes/objects/d4/d8ceb2cd332eb46de5668b5290fc6c8a692cfeb43e2451ab9016c5c071950b
```

```
krithika_s@88U9592OG990I:~/PES2UG24CS238-pes-vcs$ xxd .pes/objects/65/df4fd62f3b96e93f33b3b3924c21ebe5331bf54770d8fa7c9a5018efaf3008
00000000: 626c 6f62 2037 3534 3400 7f45 4c46 0201  blob 7544..ELF..
00000010: 0100 0000 0000 0000 0000 0100 3e00 0100  ............>...
00000020: 0000 0000 0000 0000 0000 0000 0000 0000  ................
00000030: 0000 b818 0000 0000 0000 0000 0000 4000  ..............@.
00000040: 0000 0000 4000 1300 1200 f30f 1efa 5553  ....@........US
00000050: 4c8d 9c24 00e0 f6ff 4881 ec00 1000 0048  L..$....H......H
00000060: 830c 2400 4c39 dc75 ef48 83ec 3866 0f6f  ..$.L9.u.H..8f.o
00000070: 0500 0000 0048 bb52 4541 444d 452e 6d48  .....H.README.mH
00000080: bd62 7569 6c64 2e73 6864 488b 0425 2800  .build.shdH..%(.
00000090: 0000 4889 8424 2820 0900 31c0 0f11 4424  ..H..$( ..1...D$
000000a0: 14b8 6400 0000 4889 e60f 1144 2424 488d  ..d...H....D$$H.
000000b0: 7c24 1066 0f6f 0500 0000 0048 8d54 2408  |$.f.o.....H.T$.
000000c0: c784 2410 9004 0003 0000 000f 1184 2438  ..$..........$8
000000d0: 0100 000f 1184 2448 0100 0066 0f6f 0500  ......$H...f.o..
000000e0: 0000 00c7 4424 10a4 8100 0048 895c 2434  ....D$.....H.\$4
000000f0: 6689 4424 3cc7 8424 3401 0000 0040 0000  f.D$<..$4....@..
00000100: c784 2458 0100 0073 7263 00c7 8424 5802  ..$X...src...$X.
00000110: 0000 ed81 0000 4889 ac24 7c02 0000 c684  ......H..$|.....
00000120: 2484 0200 0000 0f11 8424 5c02 0000 0f11  $........$\.....
00000130: 8424 6c02 0000 e800 0000 0085 c00f 855f  .$l............_
00000140: 0200 0048 833c 2400 0f84 3502 0000 488b  ...H.<$...5...H.
00000150: 5424 0848 85d2 0f84 0802 0000 488d 3500  T$.H........H.5.
00000160: 0000 00bf 0200 0000 31c0 e800 0000 0048  ........1......H
00000170: 8b74 2408 488b 3c24 488d 9424 2090 0400  .t$.H.<$H..$ ...
00000180: e800 0000 0085 c00f 85b8 0100 0083 bc24  ..............$
00000190: 2020 0900 030f 858b 0100 0048 399c 2444    ........H9.$D
000001a0: 9004 0074 2548 8d0d 0000 0000 ba2f 0000  ...t%H......./..
000001b0: 0048 8d35 0000 0000 488d 3d00 0000 00e8  .H.5....H.=.....
000001c0: 0000 0000 660f 1f44 0000 6683 bc24 4c90  ....f..D..f..$L.
000001d0: 0400 6475 d048 39ac 2468 9104 0074 2348  ..du.H9.$h...t#H
000001e0: 8d0d 0000 0000 ba30 0000 0048 8d35 0000  .......0...H.5..
000001f0: 0000 488d 3d00 0000 00e8 0000 0000 0f1f  ..H.=...........
00000200: 4000 80bc 2470 9104 0000 75d3 81bc 248c  @...$p....u...$.
00000210: 9204 0073 7263 0074 2148 8d0d 0000 0000  ...src.t!H......
00000220: ba31 0000 0048 8d35 0000 0000 488d 3d00  .1...H.5....H.=.
00000230: 0000 00e8 0000 0000 6690 81bc 2420 9004  ........f...$ ..
00000240: 00a4 8100 000f 85bc 0000 0081 bc24 4491  .............$D.
00000250: 0400 ed81 0000 0f85 8901 0000 81bc 2468  ............$h
00000260: 9204 0000 4000 000f 8559 0100 0048 8b84  ....@....Y...H..
00000270: 2424 9004 0048 8b94 242c 9004 0048 3344  $$...H..$,...H3D
00000280: 2414 4833 5424 1c48 09d0 7426 488d 0d00  $.H3T$.H..t&H...
00000290: 0000 00ba 3900 0000 488d 3500 0000 0048  ....9...H.5....H
000002a0: 8d3d 0000 0000 e800 0000 000f 1f80 0000  .=............
```

3a

```
krithika_s@88U9592OG990I:~/PES2UG24CS238-pes-vcs$ ./pes init
echo "hello" > file1.txt
echo "world" > file2.txt
./pes add file1.txt file2.txt
./pes status
Initialized empty PES repository in .pes/
Staged changes:
  staged:     file1.txt
  staged:     file2.txt

Unstaged changes:
  (nothing to show)

Untracked files:
  untracked:  .gitignore
  untracked:  test_objects.c
  untracked:  tree.c
  untracked:  test_objects
  untracked:  pes.c
  untracked:  commit.h
  untracked:  index.h
  untracked:  commit.c
  untracked:  test_tree
  untracked:  test_tree.c
  untracked:  README.md
  untracked:  object.c
  untracked:  Makefile
  untracked:  test_sequence.sh
  untracked:  tree.h
  untracked:  pes.h
  untracked:  index.c
```

```
krithika_s@88U9592OG990I:~/PES2UG24CS238-pes-vcs$ ./pes add file1.txt file2.txt
krithika_s@88U9592OG990I:~/PES2UG24CS238-pes-vcs$ ./pes status
Staged changes:
  staged:     file1.txt
  staged:     file2.txt

Unstaged changes:
  (nothing to show)

Untracked files:
  untracked:  .gitignore
  untracked:  test_objects.c
  untracked:  tree.c
  untracked:  test_objects
  untracked:  pes.c
  untracked:  commit.h
  untracked:  index.h
  untracked:  commit.c
  untracked:  test_tree
  untracked:  test_tree.c
  untracked:  README.md
  untracked:  object.c
  untracked:  Makefile
  untracked:  test_sequence.sh
  untracked:  tree.h
  untracked:  pes.h
  untracked:  index.c

krithika_s@88U9592OG990I:~/PES2UG24CS238-pes-vcs$
```

3b

```
krithika_s@88U9592OG990I:~/PES2UG24CS238-pes-vcs$ cat .pes/index
100644 2cf8d83d9ee29543b34a87727421fdecb7e3f3a183d337639025de576db9ebb4 1776443965 6 file1.txt
100644 e00c50e16a2df38f8d6bf809e181ad0248da6e6719f35f9f7e65d6f606199f7f 1776443965 6 file2.txt
```

4a

```
krithika_s@88U9592OG990I:~/PES2UG24CS238-pes-vcs$ ./pes log
commit 5df21d2cc4be79af197e44167eb131688d98f9c77dae5aba8e381a20512027d9
Author: PES User <pes@localhost>
Date:   1776445570

    Add farewell


commit faa8239e9965524e41104d2e01c4699a169a858a65dbd76f18feb21ea9459090
Author: PES User <pes@localhost>
Date:   1776445555

    Add world
ll


commit acb78445564c5ed9d182330bea8fc027304964e15914deaceb7bb22618fe35ff
Author: PES User <pes@localhost>
Date:   1776445533

    Initial commit


commit 4e00bf1c79aeace2a31aeb33fe800af4a5667b8e3ae9ecb63ab27dc6e3f90851
Author: PES User <pes@localhost>
Date:   1776445356

    Initial commit
ial commiQ

krithika_s@88U9592OG990I:~/PES2UG24CS238-pes-vcs$
```

4b

```
krithika_s@88U9592OG990I:~/PES2UG24CS238-pes-vcs$ find .pes -type f | sort
.pes/HEAD
.pes/index
.pes/objects/05/c8d777d0be85748b6788d1b5caf7d69754e6e68060c265ca21cbd6c05e5f43
.pes/objects/0d/36b699f2fb606b6af430b611f65c351183b9ea05c0bf82b7c5cebb6d4899b0
.pes/objects/10/d488e897a9291aca9f7f55db526cf1c0a4d2a6274de8d6729d8da036cce9fd
.pes/objects/25/ef1fa07ea68a52f800dc80756ee6b7ae34b337afedb9b46a1af8e11ec4f476
.pes/objects/2a/594d39232787fba8eb7287418aec99c8fc2ecdaf5aaf2e650eda471e566fcf
.pes/objects/2c/f8d83d9ee29543b34a87727421fdecb7e3f3a183d337639025de576db9ebb4
.pes/objects/30/1784349754fedc835be3ca066dbfd5c16cb1ffbf6539e86be514ba37fe6bf7
.pes/objects/34/3d34bf690fefa64dc0d95a01280df0e5d489c8eced213340be03ea5ffacc2b
.pes/objects/43/72fc14cf3990ffbc0d87e98d84d4cd14e07bf8547caca89d8c9f7f8f3103f4
.pes/objects/48/1407f7237f7bd749dca31e2d8422bafe211741d5556f007e0faf821ffc7179
.pes/objects/4e/00bf1c79aeace2a31aeb33fe800af4a5667b8e3ae9ecb63ab27dc6e3f90851
.pes/objects/5d/f21d2cc4be79af197e44167eb131688d98f9c77dae5aba8e381a20512027d9
.pes/objects/60/459622a6a99c40ddfcd668e7cbd0d70a4d53aeb448a251aa982ea2043e3838
.pes/objects/62/19dea03b95f88014daaf8d0c1f2c3876a4df7e0dd8f7bf4a2b8f9c4025f4e3
.pes/objects/62/27d26dfb567be18fd00cb3d22f14359e1e658f05659ea73b19aa54fcff0614
.pes/objects/65/df4fd62f3b96e93f33b3b3924c21ebe5331bf54770d8fa7c9a5018efaf3008
.pes/objects/66/224663d23e6f4d9de9e2c7e6d8764305a92a3830a1a52d3d5f4aa8007b5c39
.pes/objects/6b/1a89bbbd6a604d675a2fa7e4d59aa0d38e3867805722b60e37aa21bb195414
.pes/objects/6e/2ece108ec707b076dc85e5691d8d82601dc9d5e4b275d26d30eadb8ccdda81
.pes/objects/8c/66283a8bc77d49cd1603623f437bdd1d363a4bbf5cd5f13e10d71669c592d8
.pes/objects/8e/128fab627eae587dffd5b1bf4bc8e5a7a75af7c3aac302fc408cb35f022b5a
.pes/objects/8f/c13b3ff597c5d58c22d59cc057f3f2d5d4becafd8578835da61dbd0dd4e0b3
.pes/objects/94/4f06dc83e3ca6e0f4573268f03de16d5a3dfd2cc43165878fe1d3e6972ee35
.pes/objects/a0/9cd29b4a4df5ae81a1da9f06d60998bf142b62356158f9d991366b92fe90e5
.pes/objects/a1/6c9a05468adae6e754a3f5bf49533153c3b0f32c2444b615c142e18fbc8e35
.pes/objects/a7/4c8d33a8cdae4949c89c9996cf1d246bbd04c40826d9dc7d25fbe611236b73
.pes/objects/ac/b78445564c5ed9d182330bea8fc027304964e15914deaceb7bb22618fe35ff
.pes/objects/b9/9f25b5b7ce8abc32c8255ce7657c1a9f0f68e48f24f2ef9bcb9cda94bac19f
.pes/objects/c6/53a193f484b05908d36c4c314a6f767241aee29be344d2abaf62de28958d9d
.pes/objects/c7/727bb3a915f59a2c4543394625c111bfb925e7975d4fa9339bc24377bd9c38
.pes/objects/c9/ed899783ff70cac61dbbc4e205d04714402c853c4e5f6107d2c8b21cd3a8f4
.pes/objects/d4/d8ceb2cd332eb46de5668b5290fc6c8a692cfeb43e2451ab9016c5c071950b
.pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
.pes/objects/e0/0c50e16a2df38f8d6bf809e181ad0248da6e6719f35f9f7e65d6f606199f7f
.pes/objects/e1/8f77c5c81b723ce4688bc7e2dc8a398bfa2501649445a8d1a21bacad85fbd9
.pes/objects/e2/d8026a766be394788e227b71b9fb17cfeb33c2966577742770d2ca9e977028
.pes/objects/e9/5829cbebfc4ff105b206ece482623e221adfa1973ba6a858531b6d74739708
.pes/objects/f7/7c03788f0e18ff8b43c8134f449b8f19b890cc74faf3329e4c9bb978aefc03
.pes/objects/fa/a8239e9965524e41104d2e01c4699a169a858a65dbd76f18feb21ea9459090
.pes/refs/heads/main
krithika_s@88U9592OG990I:~/PES2UG24CS238-pes-vcs$
```

4c

```
.pes/refs/heads/main
krithika_s@88U9592OG990I:~/PES2UG24CS238-pes-vcs$ cat .pes/refs/heads/main
5df21d2cc4be79af197e44167eb131688d98f9c77dae5aba8e381a20512027d9
krithika_s@88U9592OG990I:~/PES2UG24CS238-pes-vcs$ cat .pes/HEAD
ref: refs/heads/main
krithika_s@88U9592OG990I:~/PES2UG24CS238-pes-vcs$
```

Final part:

```
krithika_s@88U9592OG990I:~/PES2UG24CS238-pes-vcs$ make test-integration
=== Running integration tests ===
bash test_sequence.sh
=== PES-VCS Integration Test ===

--- Repository Initialization ---
Initialized empty PES repository in .pes/
PASS: .pes/objects exists
PASS: .pes/refs/heads exists
PASS: .pes/HEAD exists

--- Staging Files ---
Status after add:
Staged changes:
  staged:     file.txt
  staged:     hello.txt

Unstaged changes:
  (nothing to show)

Untracked files:
  (nothing to show)


--- First Commit ---
Committed: 335b3b22221a... Initial commit

Log after first commit:
commit 335b3b22221a435f4268604786797d2d388c786a14622582f29011de684eda11
Author: PES User <pes@localhost>
Date:   1776447439

    Initial commit



--- Second Commit ---
Committed: df4ef43f1cd9... Update file.txt

--- Third Commit ---
Committed: 9e2b4e7348e5... Add farewell

--- Full History ---
commit 9e2b4e7348e59a6c1dd87711af95534d227167a30af8c5b26704d27a52d2d66c
Author: PES User <pes@localhost>
Date:   1776447439

    Add farewell
```

```
commit df4ef43f1cd9971c3e7769751135779f998e5c0fac66830b8a12cfac75bffcc2
Author: PES User <pes@localhost>
Date:   1776447439

    Update file.txt
!

commit 335b3b22221a435f4268604786797d2d388c786a14622582f29011de684eda11
Author: PES User <pes@localhost>
Date:   1776447439

    Initial commit
ial commiQ


--- Reference Chain ---
HEAD:
ref: refs/heads/main
refs/heads/main:
9e2b4e7348e59a6c1dd87711af95534d227167a30af8c5b26704d27a52d2d66c

--- Object Store ---
Objects created:
10
.pes/objects/0b/d69098bd9b9cc5934a610ab65da429b525361147faa7b5b922919e9a23143d
.pes/objects/10/1f28a12274233d119fd3c8d7c7216054ddb5605f3bae21c6fb6ee3c4c7cbfa
.pes/objects/33/5b3b22221a435f4268604786797d2d388c786a14622582f29011de684eda11
.pes/objects/58/a67ed1c161a4e89a110968310fe31e39920ef68d4c7c7e0d6695797533f50d
.pes/objects/9e/2b4e7348e59a6c1dd87711af95534d227167a30af8c5b26704d27a52d2d66c
.pes/objects/ab/5824a9ec1ef505b5480b3e37cd50d9c80be55012cabe0ca572dbf959788299
.pes/objects/b1/7b838c5951aa88c09635c5895ef7e08f7fa1974d901ce282f30e08de0ccd92
.pes/objects/d0/7733c25d2d137b7574be8c5542b562bf48bafeaa3829f61f75b8d10d5350f9
.pes/objects/db/07a1451ca9544dbf66d769b505377d765efae7adc6b97b75cc9d2b3b3da6ff
.pes/objects/df/4ef43f1cd9971c3e7769751135779f998e5c0fac66830b8a12cfac75bffcc2

=== All integration tests completed ===
krithika_s@88U9592OG990I:~/PES2UG24CS238-pes-vcs$
```

5.1

Answer:- In a version control system like PES-VCS branches are used to maintain different lines of development. Each branch represents a snapshot of the project at a certain point in history, allowing multiple independent workflows. Switching between branches means changing the entire project state to match a different commit history.A branch is basically just a file inside .pes/refs/heads/ that stores a

commit hash. So when we run pes checkout <branch>, the first step is to read

that hash from the branch file and update .pes/HEAD so it now points to that

branch. After that, the system loads the corresponding commit object and gets

the tree stored inside it. Using that tree, the working directory is updated to match

the selected branch. This means existing files may be replaced, some files may

be deleted, and new ones may be created depending on the differences. The

index also needs to be updated so that it reflects this new state. The reason this

operation is tricky is because we are directly modifying the user's files, so we must be careful not to accidentally overwrite important changes. Also, different branches can have completely different directory structures, which makes handling all cases a bit complex.

5.2

To check if the working directory is "dirty," we compare the current files with what is stored in the index. The index keeps the last staged version of each file using its hash and metadata. So for each tracked file, we read its current content, compute its hash, and compare it with the hash stored in the index. If they don't match, it means the file has been modified in the working directory.

We also use file metadata like modification time (mtime) and file size as additional quick checks before recomputing hashes, which improves efficiency. Then we also check what version of that file exists in the branch we want to switch to by comparing it with the commit tree snapshot.

If the file is modified locally and also different in the target branch, then we have a conflict situation. In that case, checkout should not proceed because it may overwrite the user's uncommitted changes and cause data loss. So the system simply stops the operation and asks the user to commit or stash the changes first before switching branches.

5.3Answer:-A detached HEAD happens when .pes/HEAD contains a commit hash directly instead of pointing to a branch. In this state, we can still make commits normally, but those commits are not linked to any branch. That means there is no permanent reference to them. Over time, if nothing points to those commits, they can be lost during garbage collection. However, they are not immediately gone. The user can still recover them by creating a new branch using that commit hash. Once a branch is created, the commit becomes safe again because it now has a reference.

6.1

To clean up unused objects, we first need to find which ones are still being used. This is done by starting from all branch heads and following their commit history. Every commit we visit is marked as reachable. From each commit, we also follow its tree and any parent commits. Then for each tree, we go through all its entries and mark the related

blobs and subtrees. To keep track of everything efficiently, we can use a hash set to store all visited object hashes.

After this process, any object that is not in this set is considered unreachable and can be deleted. In a large repository with around 100,000 commits, we might end up visiting a few hundred thousand objects in total when we include trees and blobs. This traversal ensures that only valid and referenced objects remain in the repository. It also helps in reducing disk usage by removing redundant or orphaned data.

6.2

Running garbage collection at the same time as a commit can cause race conditions. During a commit, new objects like blobs, trees, and commits are created first, and only later is the branch updated to point to the new commit. If garbage collection runs before this update, it may not see these objects as reachable and can wrongly delete them. This leads to broken references and an inconsistent repository state. To avoid this, systems use atomic updates, file locking, and temporary files so that objects are only deleted when they are fully and safely referenced.